TABLE II
OTHER NON-INVASIVE TECHNIQUES AND RESULT ANALYSIS

Technique	Principle of determination	Result and analysis
NIR spectroscopy	Absorption or emission data in 0.7-2.5μm region was compared to known data for BG. Test sites were ear lobes, finger webs, finger cuticles, forearm skin and lip mucosa (λ-1000nm to 2500nm).	NIR diffuse reflectance performed on fingers and cuticles. Good correlation with BG, 90 per cent of predictions are clinically acceptable.
Raman spectroscopy	Raman spectroscopy measures scattered light that has been influenced by the oscillation and rotation of the scatter. Technique uses laser light to induce molecular emission. Analytes are blood, water, serum and plasma.	Analytical limitations are instability in laser wavelength, intensity, errors due to other chemicals in tissue sample, and long spectral acquisition time.
Photo-acoustic spectroscopy	Photo-acoustic spectroscopy uses an optical beam to rapidly heat the sample and generate an acoustic pressure wave that can be measured by a microphone. Analytes like blood tissue phantoms and humans showed greater sensitivity.	Excellent correlation between photo-acoustic signal and BG levels shown on index fingers. However, instrumentation is currently custom-made, environmentally-sensitive, has chemical and physical interferences and is expensive.
Scatter changes	Light scattering-based measurement monitors the changes in tissue reduced scattering coefficient. Analytes are tissue phantoms and humans. Glucose measurement is proportional to the refractive index of the analyte.	Abdomens of diabetic patients show excellent correlation between scatter signal and BG. But, measurement contributes to natural physiological scattering drift.
IR spectroscopy	Epidermal surface from 10μm - 50μm enables IR measurements. (λ-700 to 1000nm).	Measurement of glucose in non-blood-containing tissues, such as mucosa. But, interferences due to albumin, red cells, gamma globulin, pH and temperature changes.
Mid-IR spectroscopy (TES)	Thermal emission spectroscopy is the absorption or emission data in 2.5μm - 25μm region. It is tested to quantify glucose in body fluid.	Non-invasive prototype follows TES, mid-IR spectral region. It measures BG with clinically-acceptable accuracy without individual daily calibrations.

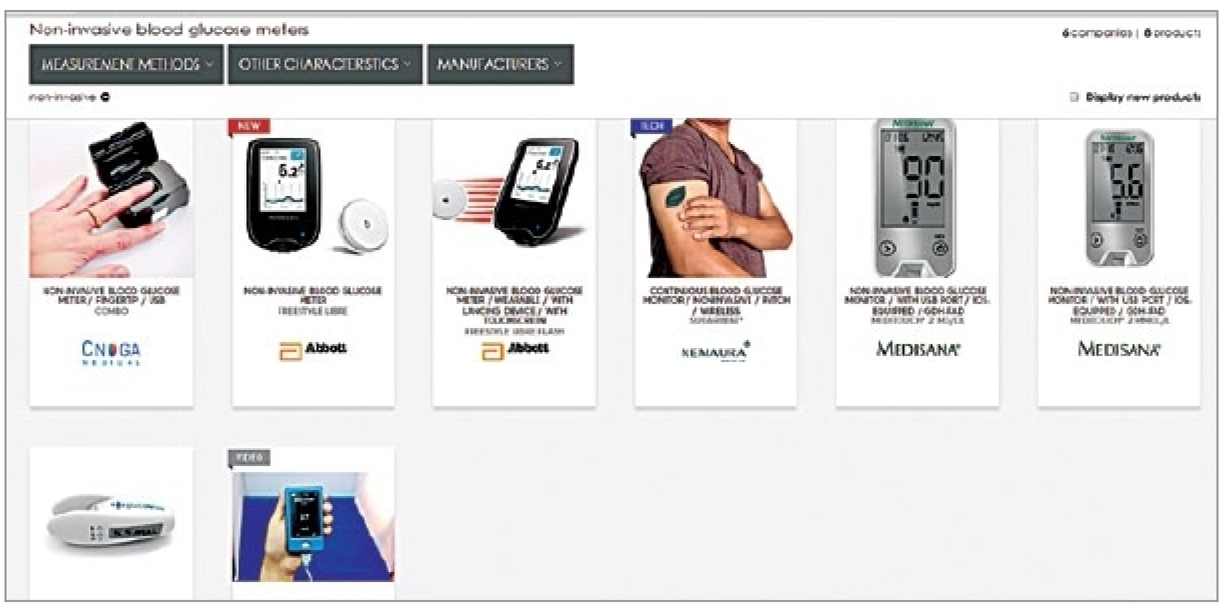

Fig. 11: Some non-invasive glucose meters of the future

non-invasive glucose meters and their usage seem imperative. Soon, indirect contact glucose meters would be feasible, too. Non-invasive BG meters by CNOGA-Combo, Abbott-Freestyle Libre, Nemaura-Sugarbeat, Medisana Glucowise and Glucotrack with continuous monitoring, USB ports, patch/wireless and touchscreen features are some non-invasive glucose meters of the future (Fig. 11). Many other meters are in their design and development phases. EFY

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